GRADE 10 / Geography / English

Atmosphere: explain heating

01 Understand

Learning goals: Distinguish incoming and outgoing radiation • Explain the natural greenhouse effect • Interpret a simple energy pathway

The Sun supplies mainly shortwave radiation. Some is reflected; some is absorbed by the atmosphere and surface.

The warmed Earth emits longwave infrared radiation. Greenhouse gases absorb and re-emit some outgoing infrared, helping warm the lower atmosphere and surface.

Albedo is reflectivity. A brighter surface generally reflects more incoming radiation, while a darker surface generally absorbs more under comparable conditions.

Simplified energy pathway

Stage	Radiation or transfer	Result
Sun → Earth	shortwave	surface absorbs some energy
Surface → atmosphere/space	longwave infrared	surface loses energy
Greenhouse gases	absorb and re-emit infrared	some energy returns towards surface

02 Follow the reasoning

Why might dark tar warm more than a pale painted surface in sunshine?

1. Compare surfaces under the same sunlight and other similar conditions.

2. Dark tar generally has a lower albedo and absorbs a larger share of incoming radiation.

3. More absorbed energy can raise its temperature; wind, moisture and material properties also affect actual temperatures.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Which has higher albedo: fresh snow or dark tar?

2. Name the radiation mainly emitted by the Earth.

3. Why is the greenhouse effect different from ozone depletion?

04 Check and correct

1. Fresh snow.

2. Longwave infrared radiation.

3. The greenhouse effect concerns infrared absorption and re-emission; ozone depletion concerns reduced protection from harmful ultraviolet radiation.

Common mistake to avoid

Do not explain global warming as the ozone hole letting in extra heat; these are different atmospheric processes.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: The atmosphere: heating of the atmosphere

DBE CAPS, Grades 10–12: Geography. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID geo-10.